

Computer Science & Information Technology

Discrete Mathematics

DPP: 2

Combinatorics

Q1 The generating function for the following sequence is

2, 4, 6, 8, 10,

(A) $s = \frac{2}{(1-x)^2}$

(B) $s = \frac{2}{(1+x)^2}$

(C) $s = \frac{1}{(1-x)^2}$

(D) $s = \frac{1}{(1+x)^2}$

Q2 Find coefficient of x^2y^3 in the expansion of $(2x-3y)^5$.

Q3 Find coefficient of x^6 in the expansion of $\left(2x^3 - \frac{1}{x^2}\right)^{12}$

Q4 Find coefficient of x^{15} in the expansion of $(x^2 + x^3 + x^4 + x^5)^3$

Q5 The sequence 1, 3, 7, 15, 31, 63, satisfies the recurrence relation

$$a_n = 3a_{n-1} - 2a_{n-2}$$

Find the generating function for the sequence.

(A) $S = \frac{1}{(1+3x+2x^2)}$

(B) $S = \frac{1}{(1-3x+2x^2)}$

(C) $S = \frac{1}{(1-3x-2x^2)}$

(D) $S = \frac{1}{(1+3x-2x^2)}$

Q6 Find a generating function for sequence 1, 4, 16, 64,

(A) $S = \frac{2}{1+4x}$

(B) $S = \frac{2}{1-4x}$

(C) $S = \frac{1}{1-4x}$

(D) $S = \frac{1}{1+4x}$

Q7 Find a generating function for sequence: 1, -5, 25, -125, 625, ...

(A) $\frac{1}{(1+25x)}$

(B) $\frac{1}{(1-25x)}$

(C) $\frac{1}{(1-5x)}$

(D) $\frac{1}{(1+5x)}$

Q8 Find the closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 4(2)^n + 5(-4)^n$, for all $n = 0, 1, 2, 3, \dots$

(A) $\frac{4}{(1-2x)} + \frac{5}{(1+4x)}$

(B) $\frac{4}{(1-2x)} - \frac{5}{(1+4x)}$

(C) $\frac{4}{(1+2x)} + \frac{5}{(1+4x)}$

(D) $\frac{4}{(1-2x)} + \frac{5}{(1-4x)}$

Q9 $a_n = 3a_{n-1} - 2a_{n-2}$, $n \geq 2$ with initial terms $a_0 = 1$ and $a_1 = 3$.

Solution of above recurrence relation using generating function is

(A) $2^{n+1} - 1$

(B) $2^{n+1} + 1$

(C) $2^{n-1} - 1$

(D) $2^{n-1} + 1$

Q10 If the generating function of the sequence $\{a_0, a_1, a_2, \dots\}$ is $\frac{x}{(1-x^3)^2}$ then $(a_1 + a_2)$ is equal to?

Answer Key

Q1 (A)

Q2 -1080~-1080

Q3 59136~59136

Q4 1~1

Q5 (B)

Q6 (C)

Q7 (D)

Q8 (A)

Q9 (A)

Q10 3~3



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Hints & Solutions

Q1 Text Solution:

$$\begin{aligned}
 S &= 2 + 4x + 6x^2 + 8x^3 + 10x^4 + \dots \\
 \Rightarrow xS &= 2x + 4x^2 + 6x^3 + 8x^4 + 10x^5 + \dots \\
 \Rightarrow S - xS &= 2 + 2x + 2x^2 + 2x^3 + \dots \\
 \Rightarrow S(1-x) &= 2(1+x+x^2+x^3+\dots) \\
 \Rightarrow S(1-x) &= \frac{2}{1-x} \\
 \Rightarrow S &= \frac{2}{(1-x)^2}
 \end{aligned}$$

Q2 Text Solution:

$$\begin{aligned}
 (2x - 3y)^5 &= \sum_{k=0}^5 {}^5C_k (2x)^{5-k} (-3y)^k \\
 \text{The term for } x^2y^3 &\text{ is } \\
 {}^5C_3 (2x)^{5-3} (-3y)^3 & \\
 = \left(\frac{5 \times 4 \times 3!}{2 \times 3!} \right) \times (4x^2) \times (-27y^3) & \\
 = -1080 x^2y^3 &
 \end{aligned}$$

Q3 Text Solution:

$$\begin{aligned}
 &= \left(\frac{2x^5 - 1}{x^2} \right)^{12} \\
 &= \frac{1}{x^{24}} (2x^5 - 1)^{12} \\
 \text{So, now,} & \\
 (2x^5 - 1)^{12} &= \sum_{k=0}^{12} {}^{12}C_k (2x^5)^{12-k} (-1)^k \\
 \text{for } x^{30}, & \\
 = {}^{12}C_6 (2x^5)^{12-6} (-1)^6 & \\
 = {}^{12}C_6 (2x^5)^6 & \\
 = 924 \times 64 & \\
 = 59136 &
 \end{aligned}$$

Q4 Text Solution:

$$\begin{aligned}
 (x^2 + x^3 + x^4 + x^5)^3 & \\
 = (x^2)^3 (1 + x + x^2 + x^3)^3 & \\
 = x^6 \left(\frac{1-x^4}{1-x} \right)^3 & \\
 = x^6 (1-x^4)^3 (1-x)^{-3} & \\
 = x^6 (1-x^{12} - 3x^4 + 3x^8) (1-x)^{-3} & \\
 = (x^6 - x^{18} - 3x^{10} + 3x^{14}) (1-x)^{-3} & \\
 = (x^6 \times {}^{3+9-1}C_9 x^9 - 3x^{10} \times {}^{3+5-1}C_5 x^5 & \\
 + 3x^{14} \times {}^{3+1-1}C_1 x^1 & \\
 = {}^{11}C_9 x^{15} - 3 \times {}^7C_5 x^{15} + 3 \times {}^3C_1 x^{15} & \\
 = \left(\frac{11 \times 10}{2} - \frac{3 \times 7 \times 6}{2} + 3 \times 3 \right) x^{15} & \\
 = (55 - 63 + 9) x^{15} &
 \end{aligned}$$

$$= 1 x^{15}$$

Q5 Text Solution:

$$\begin{aligned}
 S &= 1 + 3x + 7x^2 + 15x^3 + 31x^4 + \dots \\
 \Rightarrow xS &= x + 3x^2 + 7x^3 + 15x^4 + \dots \\
 \Rightarrow S - xS &= 1 + 2x + 4x^2 + 8x^3 + 16x^4 + \dots \\
 \Rightarrow S - xS &= \frac{1}{1-2x} \\
 \Rightarrow S &= \frac{1}{(1-x)(1-2x)} = \frac{1}{1-3x+2x^2}
 \end{aligned}$$

Q6 Text Solution:

$$\begin{aligned}
 S &= 1 + 4x + 16x^2 + 64x^3 + \dots \\
 \Rightarrow xS &= x + 4x^2 + 16x^3 + 64x^4 + \dots \\
 \Rightarrow S - xS &= 1 + 3x + 12x^2 + 48x^3 + \dots \\
 \Rightarrow S(1-x) &= 1 + \frac{3x}{1-4x} \\
 \Rightarrow S(1-x) &= \frac{1-x}{1-4x} \\
 \Rightarrow S &= \frac{1}{1-4x}
 \end{aligned}$$

Q7 Text Solution:

$$\begin{aligned}
 S &= 1 - 5x + 25x^2 - 125x^3 + 625x^4 - \dots \\
 S &= (1 + 25x^2 + 625x^4 + \dots) - (5x + 125x^3 + \dots) \\
 S &= \frac{1}{1-25x^2} - \frac{5x}{1-25x^2} \\
 S &= \frac{1-5x}{1-25x^2} \\
 S &= \frac{(1-5x)}{(1+5x)(1-5x)} = \frac{1}{1+5x}
 \end{aligned}$$

Q8 Text Solution:

$$\begin{aligned}
 G(x) &= \sum_{n=0}^{\infty} \{4(2)^n + 5(-4)^n\} x^n \\
 &= 4 \sum_{n=0}^{\infty} 2^n x^n + 5 \sum_{n=0}^{\infty} (-4)^n x^n \\
 &= 4 \left(\frac{1}{1-2x} \right) + 5 \left(\frac{1}{1-(-4x)} \right) \\
 &= \frac{4}{1-2x} + \frac{5}{1+4x}
 \end{aligned}$$

Q9 Text Solution:

$$\begin{aligned}
 G(x) &= \sum_{n=0}^{\infty} a_n x^n \\
 &= a_0 x^0 + a_1 x^1 + \sum_{n=2}^{\infty} a_n x^n \dots (i) \\
 &= 1 + 3x + \sum_{n=2}^{\infty} (3a_{n-1} - 2a_{n-2}) x^n \\
 &= 1 + 3x + 3 \sum_{n=2}^{\infty} a_{n-1} x^n - 2 \sum_{n=2}^{\infty} a_{n-2} x^n \\
 &= 1 + 3x + 3x \sum_{n=2}^{\infty} a_{n-1} x^{n-1} - 2x^2 \sum_{n=2}^{\infty} a_{n-2} x^{n-2} \\
 &= 1 + 3x + 3x \sum_{n=1}^{\infty} a_n x^n - 2x^2 \sum_{n=0}^{\infty} a_n x^n \\
 &= 1 + 3x + 3x (a_1 x^1 + \sum_{n=2}^{\infty} a_n x^n) \\
 &\quad - 2x^2 G(x)
 \end{aligned}$$



$$= 1 + 3x + 3x(3x + G(x) - 1 - 3x) - 2x^2 G(x)$$

$$= 1 + 3x + 3xG(x) - 3x - 2x^2 G(x)$$

$$G(x) = 1 + 3xG(x) - 2x^2 G(x)$$

$$G(x) = \frac{1}{2x^2 - 3x + 1} = \frac{1}{(2x-1)(x-1)}$$

Using partial fraction decomposition,

$$\Rightarrow G(x) = \frac{1}{(2x-1)(x-1)} = \frac{-2}{(2x-1)} + \frac{1}{(x-1)}$$

$$= \frac{2}{(1-2x)} - \frac{1}{(1-x)}$$

$$\Rightarrow G(x) = 2(1+n-1)C_n 2^n - (1x1)$$

$$= 2^{n+1} - 1$$

$$\frac{x}{(1-x^3)^2} = x[1-x^3]^{-2} = x \cdot {}^{2+r-1}C_r (x^3)^r$$

$$= {}^{1+r}C_r x^{3r+1}$$

For a_1 ,

$${}^{1+0}C_0 x^{3 \times 0 + 1}$$

$$= {}^1C_0 x^1$$

$$= 1x^1$$

$$a_1 = 1$$

For a_4 ,

$$3r + 1 = 4$$

$$r = 1$$

$${}^{1+1}C_1 x^4$$

$$= 2x^4$$

$$a_4 = 2$$

$$a_1 + a_4 = 1 + 2 = 3$$

Q10 Text Solution:



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